

# Python Data Science Libraries

## Complete Functions & Methods Reference Guide

This reference covers the most important functions and methods in **NumPy**, **Pandas**, **Matplotlib**, and **Seaborn** — each with a concise one-line purpose description for quick lookup.

### NumPy Scientific Computing / Arrays

#### Array Creation

| Function / Method                | Purpose                                                     |
|----------------------------------|-------------------------------------------------------------|
| <code>np.array()</code>          | Convert a list/tuple into an ndarray.                       |
| <code>np.zeros()</code>          | Create an array filled with zeros.                          |
| <code>np.ones()</code>           | Create an array filled with ones.                           |
| <code>np.full()</code>           | Create an array filled with a given constant value.         |
| <code>np.empty()</code>          | Create an uninitialized array (random memory values).       |
| <code>np.arange()</code>         | Create evenly spaced values within a given range.           |
| <code>np.linspace()</code>       | Create evenly spaced numbers over a specified interval.     |
| <code>np.eye()</code>            | Create a 2-D identity matrix (ones on diagonal).            |
| <code>np.random.rand()</code>    | Generate random floats in [0, 1) with uniform distribution. |
| <code>np.random.randn()</code>   | Generate random samples from standard normal distribution.  |
| <code>np.random.randint()</code> | Generate random integers between low and high.              |

#### Array Manipulation

| Function / Method             | Purpose                                                 |
|-------------------------------|---------------------------------------------------------|
| <code>np.reshape()</code>     | Change shape of an array without altering its data.     |
| <code>np.flatten()</code>     | Return a copy of the array collapsed into 1-D.          |
| <code>np.ravel()</code>       | Return a contiguous 1-D array (view if possible).       |
| <code>np.transpose()</code>   | Permute array dimensions; shorthand: <code>arr.T</code> |
| <code>np.concatenate()</code> | Join a sequence of arrays along an existing axis.       |
| <code>np.stack()</code>       | Join arrays along a new axis.                           |
| <code>np.hstack()</code>      | Stack arrays in sequence horizontally (column-wise).    |
| <code>np.vstack()</code>      | Stack arrays in sequence vertically (row-wise).         |

| Function / Method             | Purpose                                    |
|-------------------------------|--------------------------------------------|
| <code>np.split()</code>       | Split an array into multiple sub-arrays.   |
| <code>np.expand_dims()</code> | Insert a new axis at a specified position. |
| <code>np.squeeze()</code>     | Remove axes of length one from the array.  |

## Math & Statistics

| Function / Method                             | Purpose                                             |
|-----------------------------------------------|-----------------------------------------------------|
| <code>np.sum()</code>                         | Sum of array elements over a given axis.            |
| <code>np.mean()</code>                        | Compute the arithmetic mean along a given axis.     |
| <code>np.median()</code>                      | Compute the median of array elements.               |
| <code>np.std()</code>                         | Compute the standard deviation.                     |
| <code>np.var()</code>                         | Compute the variance.                               |
| <code>np.min()</code> / <code>np.max()</code> | Return minimum / maximum value of array.            |
| <code>np.argmin/argmax()</code>               | Return indices of minimum / maximum values.         |
| <code>np.cumsum()</code>                      | Return cumulative sum of elements along axis.       |
| <code>np.dot()</code>                         | Dot product of two arrays (matrix multiplication).  |
| <code>np.sqrt()</code>                        | Element-wise square root of array values.           |
| <code>np.abs()</code>                         | Element-wise absolute value.                        |
| <code>np.log()</code> / <code>np.exp()</code> | Element-wise natural log / exponential.             |
| <code>np.clip()</code>                        | Limit array values to a given [min, max] range.     |
| <code>np.unique()</code>                      | Find sorted unique elements of an array.            |
| <code>np.where()</code>                       | Return elements from x or y depending on condition. |
| <code>np.percentile()</code>                  | Compute the q-th percentile of array data.          |

## Linear Algebra

| Function / Method              | Purpose                                         |
|--------------------------------|-------------------------------------------------|
| <code>np.linalg.inv()</code>   | Compute the multiplicative inverse of a matrix. |
| <code>np.linalg.det()</code>   | Compute the determinant of a matrix.            |
| <code>np.linalg.eig()</code>   | Compute eigenvalues and eigenvectors.           |
| <code>np.linalg.norm()</code>  | Compute matrix or vector norm.                  |
| <code>np.linalg.solve()</code> | Solve a linear matrix equation $Ax = b$ .       |

# Pandas Data Analysis / DataFrames

## DataFrame & Series Creation

| Function / Method            | Purpose                                                   |
|------------------------------|-----------------------------------------------------------|
| <code>pd.DataFrame()</code>  | Create a 2-D labeled data structure from dict/list/array. |
| <code>pd.Series()</code>     | Create a 1-D labeled array capable of holding any type.   |
| <code>pd.read_csv()</code>   | Read a CSV file into a DataFrame.                         |
| <code>pd.read_excel()</code> | Read an Excel file into a DataFrame.                      |
| <code>pd.read_json()</code>  | Read a JSON string/file into a DataFrame.                 |
| <code>pd.read_sql()</code>   | Read SQL query or table into a DataFrame.                 |
| <code>df.to_csv()</code>     | Write DataFrame to a comma-separated values file.         |
| <code>df.to_excel()</code>   | Write DataFrame to an Excel spreadsheet.                  |

## Exploration & Info

| Function / Method              | Purpose                                              |
|--------------------------------|------------------------------------------------------|
| <code>df.head(n)</code>        | Return the first n rows of the DataFrame.            |
| <code>df.tail(n)</code>        | Return the last n rows of the DataFrame.             |
| <code>df.info()</code>         | Print concise summary: dtypes, memory, null counts.  |
| <code>df.describe()</code>     | Generate descriptive statistics for numeric columns. |
| <code>df.shape</code>          | Return a tuple (rows, cols) of the DataFrame.        |
| <code>df.dtypes</code>         | Return the data type of each column.                 |
| <code>df.columns</code>        | Return the column label Index object.                |
| <code>df.index</code>          | Return the row label Index object.                   |
| <code>df.value_counts()</code> | Return counts of unique values in a Series.          |

## Selection & Filtering

| Function / Method             | Purpose                                           |
|-------------------------------|---------------------------------------------------|
| <code>df['col']</code>        | Select a single column by label (returns Series). |
| <code>df[['c1', 'c2']]</code> | Select multiple columns by label list.            |
| <code>df.loc[]</code>         | Label-based indexing for rows and columns.        |
| <code>df.iloc[]</code>        | Integer position-based indexing.                  |
| <code>df.at[]</code>          | Access a single scalar value by label.            |

| Function / Method         | Purpose                                             |
|---------------------------|-----------------------------------------------------|
| <code>df.iat[]</code>     | Access a single scalar value by integer position.   |
| <code>df.query()</code>   | Filter rows using a query string expression.        |
| <code>df.isin()</code>    | Check each element against a list of values.        |
| <code>df.between()</code> | Check if Series values fall within a numeric range. |

## Cleaning & Transformation

| Function / Method                              | Purpose                                                |
|------------------------------------------------|--------------------------------------------------------|
| <code>df.dropna()</code>                       | Remove rows/columns containing missing (NaN) values.   |
| <code>df.fillna()</code>                       | Fill missing values with a scalar, method, or dict.    |
| <code>df.isna()</code> / <code>isnull()</code> | Return boolean mask of missing values.                 |
| <code>df.drop()</code>                         | Drop specified rows or columns by label.               |
| <code>df.rename()</code>                       | Rename columns or index labels with a mapping.         |
| <code>df.astype()</code>                       | Cast a column or DataFrame to a specified dtype.       |
| <code>df.replace()</code>                      | Replace values with other values element-wise.         |
| <code>df.apply()</code>                        | Apply a function along an axis of the DataFrame.       |
| <code>df.map()</code>                          | Map values of a Series through a function or dict.     |
| <code>df.str.*</code>                          | Vectorized string methods (strip, split, contains...). |
| <code>df.duplicated()</code>                   | Return boolean Series marking duplicate rows.          |
| <code>df.drop_duplicates()</code>              | Remove duplicate rows from the DataFrame.              |
| <code>df.sort_values()</code>                  | Sort DataFrame by one or more column values.           |
| <code>df.reset_index()</code>                  | Reset the index to default integer range.              |
| <code>df.set_index()</code>                    | Set a column as the DataFrame row index.               |

## Aggregation & Grouping

| Function / Method             | Purpose                                             |
|-------------------------------|-----------------------------------------------------|
| <code>df.groupby()</code>     | Group rows by column values for aggregation.        |
| <code>df.agg()</code>         | Aggregate using one or more named functions.        |
| <code>df.pivot_table()</code> | Create a spreadsheet-style pivot table.             |
| <code>df.merge()</code>       | SQL-style join of two DataFrames on keys.           |
| <code>df.join()</code>        | Join DataFrame with another on index or key column. |
| <code>pd.concat()</code>      | Concatenate DataFrames along rows or columns.       |
| <code>df.melt()</code>        | Unpivot a DataFrame from wide to long format.       |

| Function / Method          | Purpose                                                |
|----------------------------|--------------------------------------------------------|
| <code>df.pivot()</code>    | Reshape data using values from columns as keys.        |
| <code>df.crosstab()</code> | Compute cross-tabulation (frequency table) of factors. |

## Figure & Axes Setup

| Function / Method               | Purpose                                                     |
|---------------------------------|-------------------------------------------------------------|
| <code>plt.figure()</code>       | Create a new figure with optional size/DPI settings.        |
| <code>plt.subplots()</code>     | Create a figure and a grid of Axes subplots.                |
| <code>plt.subplot()</code>      | Add a single subplot to the current figure.                 |
| <code>fig.add_axes()</code>     | Add Axes to the figure at specific [left,bot,w,h] position. |
| <code>plt.tight_layout()</code> | Automatically adjust subplot params to fit the figure.      |
| <code>plt.savefig()</code>      | Save the current figure to a file (PNG, PDF, SVG...).       |
| <code>plt.show()</code>         | Display all open figures.                                   |
| <code>plt.close()</code>        | Close a figure window or all figures.                       |

## Basic Plot Types

| Function / Method               | Purpose                                                            |
|---------------------------------|--------------------------------------------------------------------|
| <code>plt.plot()</code>         | Plot y vs x as lines and/or markers.                               |
| <code>plt.scatter()</code>      | Make a scatter plot of x vs y with optional color/size.            |
| <code>plt.bar()</code>          | Make a vertical bar chart.                                         |
| <code>plt.barh()</code>         | Make a horizontal bar chart.                                       |
| <code>plt.hist()</code>         | Plot a histogram of data distribution.                             |
| <code>plt.pie()</code>          | Draw a pie chart showing proportions.                              |
| <code>plt.boxplot()</code>      | Draw box-and-whisker plots for statistical summaries.              |
| <code>plt.violinplot()</code>   | Draw violin plots showing data distribution & probability.         |
| <code>plt.stem()</code>         | Plot vertical lines (stems) from baseline to data points.          |
| <code>plt.step()</code>         | Make a step plot connecting points with horizontal/vertical lines. |
| <code>plt.fill_between()</code> | Fill the area between two horizontal curves.                       |

## Labels, Legends & Formatting

| Function / Method         | Purpose                            |
|---------------------------|------------------------------------|
| <code>plt.title()</code>  | Set the title of the current axes. |
| <code>plt.xlabel()</code> | Set the label for the x-axis.      |
| <code>plt.ylabel()</code> | Set the label for the y-axis.      |

| Function / Method                                 | Purpose                                             |
|---------------------------------------------------|-----------------------------------------------------|
| <code>plt.legend()</code>                         | Place a legend on the axes.                         |
| <code>plt.xlim()</code> / <code>ylim()</code>     | Set x-axis / y-axis display limits.                 |
| <code>plt.xticks()</code> / <code>yticks()</code> | Set tick locations and labels on x/y axis.          |
| <code>plt.grid()</code>                           | Turn the axes grid on or off.                       |
| <code>plt.annotate()</code>                       | Annotate a point on the plot with text + arrow.     |
| <code>plt.text()</code>                           | Add text string at arbitrary axes coordinates.      |
| <code>plt.colorbar()</code>                       | Add a colorbar to a mappable image or contour plot. |
| <code>ax.set_aspect()</code>                      | Set the aspect ratio of the axes.                   |
| <code>ax.invert_xaxis/yaxis()</code>              | Invert the direction of the x or y axis.            |

## Advanced / Statistical Plots

| Function / Method                                   | Purpose                                                |
|-----------------------------------------------------|--------------------------------------------------------|
| <code>plt.contour()</code>                          | Draw contour lines of a 3-D surface.                   |
| <code>plt.contourf()</code>                         | Draw filled contour regions of a 3-D surface.          |
| <code>plt.imshow()</code>                           | Display image data or a 2-D array as an image.         |
| <code>plt.quiver()</code>                           | Plot a 2-D field of arrows (vector fields).            |
| <code>plt.errorbar()</code>                         | Plot y vs x with error bars on y (and/or x).           |
| <code>plt.hlines()</code> / <code>vlines()</code>   | Draw horizontal / vertical lines across the axes.      |
| <code>plt.axhline()</code> / <code>axvline()</code> | Add infinite horizontal / vertical line at a constant. |
| <code>plt.pcolormesh()</code>                       | Create a pseudo-color plot of a 2-D array.             |
| <code>mpl3d.Axes3D.plot_surface()</code>            | Draw a 3-D surface plot with Axes3D toolkit.           |

# Seaborn Statistical Data Visualization

## Distribution Plots

| Function / Method           | Purpose                                                  |
|-----------------------------|----------------------------------------------------------|
| <code>sns.histplot()</code> | Plot a histogram (univariate or bivariate distribution). |
| <code>sns.kdeplot()</code>  | Plot a kernel density estimate of data distribution.     |
| <code>sns.ecdfplot()</code> | Plot an empirical cumulative distribution function.      |
| <code>sns.rugplot()</code>  | Draw small ticks along axes to show data locations.      |
| <code>sns.displot()</code>  | Figure-level interface for distribution plots.           |

## Categorical Plots

| Function / Method             | Purpose                                                |
|-------------------------------|--------------------------------------------------------|
| <code>sns.barplot()</code>    | Show point estimates and confidence intervals as bars. |
| <code>sns.countplot()</code>  | Show counts of observations in each categorical bin.   |
| <code>sns.boxplot()</code>    | Draw a box-and-whisker plot for numeric distributions. |
| <code>sns.violinplot()</code> | Combine box plot with kernel density on each side.     |
| <code>sns.stripplot()</code>  | Draw a scatter plot with jitter for categorical data.  |
| <code>sns.swarmplot()</code>  | Non-overlapping categorical scatter plot along axis.   |
| <code>sns.pointplot()</code>  | Show estimates and confidence intervals using markers. |
| <code>sns.catplot()</code>    | Figure-level interface for all categorical plot types. |

## Relational Plots

| Function / Method              | Purpose                                                     |
|--------------------------------|-------------------------------------------------------------|
| <code>sns.scatterplot()</code> | Draw a scatter plot using x, y, hue, size, style.           |
| <code>sns.lineplot()</code>    | Draw a line plot with optional CI bands.                    |
| <code>sns.relplot()</code>     | Figure-level interface for relational (scatter/line) plots. |

## Regression & Matrix Plots

| Function / Method            | Purpose                                                 |
|------------------------------|---------------------------------------------------------|
| <code>sns.regplot()</code>   | Plot data and a linear regression model fit.            |
| <code>sns.lmplot()</code>    | Plot data with regression fit across facets/hue groups. |
| <code>sns.residplot()</code> | Plot residuals of a linear regression model.            |
| <code>sns.heatmap()</code>   | Plot rectangular data as a color-encoded matrix.        |

| Function / Method             | Purpose                                   |
|-------------------------------|-------------------------------------------|
| <code>sns.clustermap()</code> | Plot a hierarchically clustered heat map. |

## Multi-Plot Grids

| Function / Method            | Purpose                                                 |
|------------------------------|---------------------------------------------------------|
| <code>sns.pairplot()</code>  | Plot pairwise relationships across all numeric columns. |
| <code>sns.FacetGrid()</code> | Multi-plot grid for conditioning on a variable.         |
| <code>sns.PairGrid()</code>  | Axes-level grid for pairwise relationships.             |
| <code>sns.JointGrid()</code> | Grid with joint and marginal axes for bivariate data.   |
| <code>sns.jointplot()</code> | Draw joint & marginal distributions for two variables.  |

## Theming & Aesthetics

| Function / Method                | Purpose                                                    |
|----------------------------------|------------------------------------------------------------|
| <code>sns.set_theme()</code>     | Set the aesthetic parameters in one call (style, palette). |
| <code>sns.set_style()</code>     | Set the background style: white, dark, whitegrid, etc.     |
| <code>sns.set_palette()</code>   | Set the default color palette for all subsequent plots.    |
| <code>sns.color_palette()</code> | Return a list of colors that form a seaborn palette.       |
| <code>sns.despine()</code>       | Remove top and right axis spines for a cleaner look.       |
| <code>sns.set_context()</code>   | Scale plot elements for paper/talk/poster/notebook.        |